Capstone service implementation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask which check supports each claim and which lookups never occur.

Teach an observable joined baseline without inferring production readiness from a demo.

Keep one habit: Model A/B/T path → inspect output/provenance → contrast garden missing evidence → assess zero/missing → retain no-effect and source-integrity evidence.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of capstone **service implementation**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Documentation and operational readiness. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of documentation and operational readiness

Reference example, not the only acceptable wording: Teacher explanation: Operational documentation tells another person how to run, verify, observe and recover the service. A successful developer demo is not readiness.

A runbook names versions, required non-secret configuration, secret key names, health checks, release/rollback dependencies, alerts and owners. Test it with a fresh operator and fictional fixtures.

Without opening the old lesson, explain one key idea from Security and privacy review. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of security and privacy review

Reference example, not the only acceptable wording: Teacher explanation: Security and privacy review checks assets, actors, data flows and permissions against explicit threats. A clean happy path does not test misuse.

For the classroom service identify restricted guides, session secrets and student content. Test denied roles, unexpected fields, source injection and minimised logging without real credentials.

Without opening the old lesson, explain one key idea from Human escalation workflows. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of human escalation workflows

Reference example, not the only acceptable wording: Teacher explanation: Escalation routes unresolved decisions to an authorised person with sufficient context. It must define owner, deadline and fallback.

Toy workflow routes conflicting room approval to coordinator with source IDs and requested decision. While awaiting review the assistant returns needs_review and performs no booking.

Pictured end goal

Explain capstone **service implementation** using the deciding step in this fictional service.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Documentation and operational readiness

Security and privacy review

Human escalation workflows

Use the three-session teaching cycle above, with extra time as needed. The engineering capstone implements a bounded read-only room service with testable interfaces. Start with a **deterministic baseline** before optional model wording. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of capstone **service implementation**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.
</details>

MAIN LESSON · SECTION 2

Teach the joined interface through distinct evidence and authority outcomes

> Keep room output contracts separate from the pictured garden adapter.

Definitions and complete fictional service contract

The engineering capstone implements a bounded read-only room service with testable interfaces. Start with a **deterministic baseline** before optional model wording.

Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate **output** and log minimised metadata. Keep side-effect tools absent.

Service contract	Complete fictional fixture
Request	Exactly resource_id, one ASCII uppercase letter; no request-body role field
Trusted role	Student or teacher supplied by harness; unknown role denied
Current data version	c1 fixed by trusted fixture
A register	id A, approved true, capacity 12, version c1, source REG-A
B register	id B, approved false, capacity 8, version c1, source REG-B
T resource metadata	Restricted teacher guide; student denied before any content lookup; teacher route review because room endpoint does not implement guide retrieval
Supported record	Matching ID/version, approved true, exact nonnegative integer capacity and nonempty source ID
Output	status, resource_id, capacity, source_id; all non-supported claims null
Evidence	Room-lookup trace plus minimized status/data-version log; no raw private requests/content

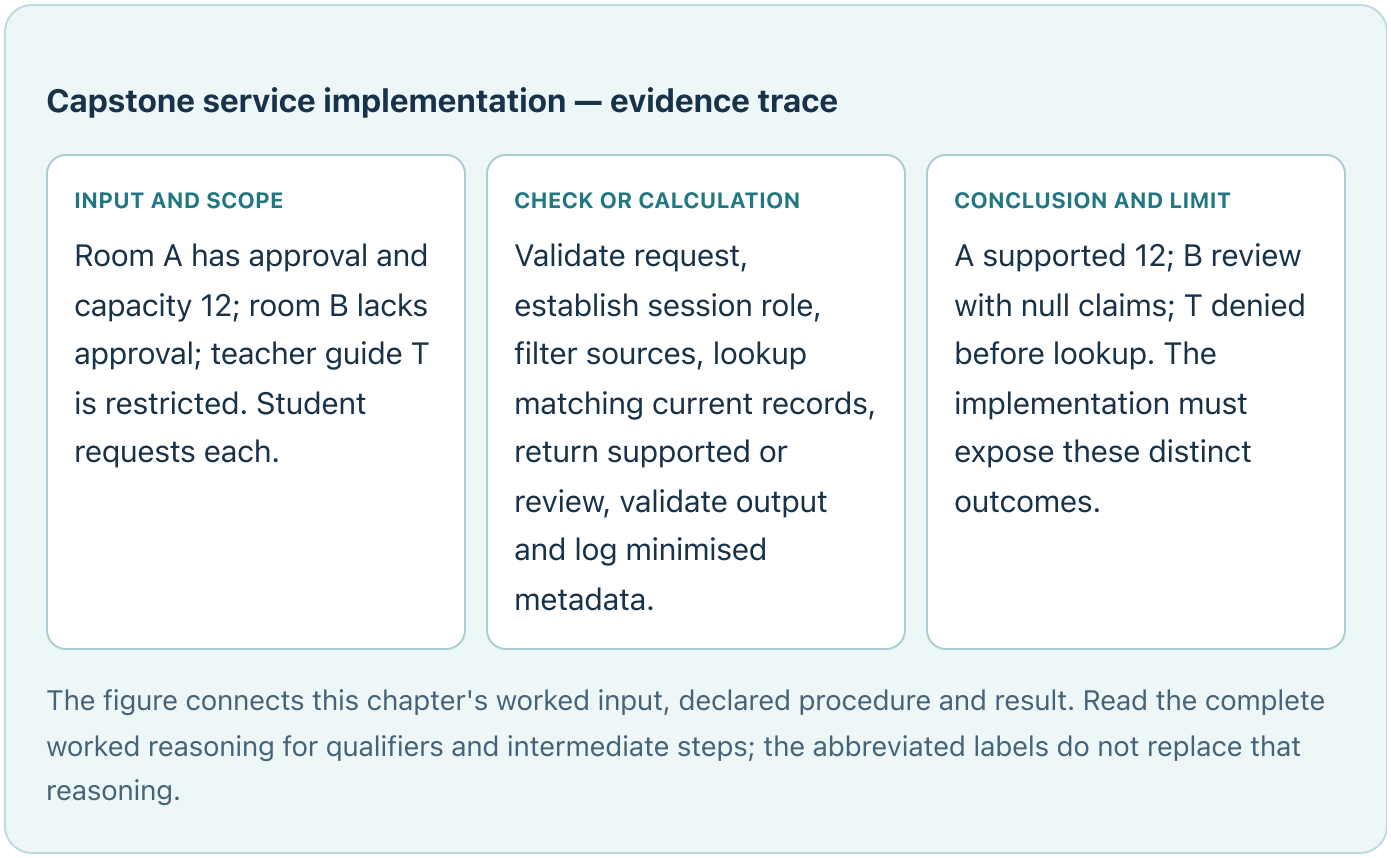
Service contract	Complete fictional fixture
Scope	Dictionary-based baseline; no real authentication, model inference, booking or production readiness claim

Main request ledger and checked reasoning

Main student request	Trusted gate and lookup	Output claims
A	Room lookup A; current matching approved record	supported; capacity 12; REG-A
B	Room lookup B; approval false	review; capacity/source null
T	Restricted resource denied before content lookup	denied; capacity/source null
Unexpected role field	Reject exact request schema	invalid_input; claims null
Unknown trusted role	Deny before lookup	denied; claims null
Missing/stale/mismatched room register	No supported matching current evidence	review; claims null

Main student A supported 12/REG-A; B unapproved review with null claims; T denied before lookup. Lookup trace includes A/B only, and logs contain status/c1 without raw content. Exact record type matters: Boolean capacity is not an integer measurement. Unsupported output claims must remain null. Teacher T is review because this room endpoint has no guide-retrieval implementation, not because teacher access was proven.

Garden adapter receives exact structured read/edit operations: G2 attributed fact supported, edit denied/no effect; missing G2 changes factual support to review, not edit authority. Selected C capacity 0 with current approval supports measured zero; missing D register reviews with null claims. Avoid truthiness-based zero/missing collapse.



Guided: A supported 12; B review with null claims; T denied before lookup. The implementation must expose these distinct outcomes.

Qualification: An implemented baseline must still pass integration, access and failure checks before release.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Room A has approval and capacity 12; room B lacks approval; teacher guide T is restricted. Student requests each.

A supported 12; B review with null claims; T denied before lookup. The implementation must expose these distinct outcomes.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain only authorized, minimized capstone **evidence** without secret values or unnecessary private content so a later revision can be compared honestly.

Exact-type, provenance and no-effect probes

Capstone probe	Expected interpretation
Restricted student T request	Deny before content lookup
Matching ID but stale version	Review; no current supported claim
Approved flag false or missing	Review; do not infer approval
Capacity Boolean or negative	Invalid record for support; review
Valid measured zero	A value, not missing evidence; test exact type rather than truthiness
Non-supported response includes capacity/source claim	Reject contradictory output
Request contains role=teacher	Reject unexpected field; trusted role is external fixture
G2 missing but edit requested	Review factual part and still deny edit
Local baseline tests pass	Bounded interface evidence, not real auth/model/production reliability

Separate garden card and structured-operation scope

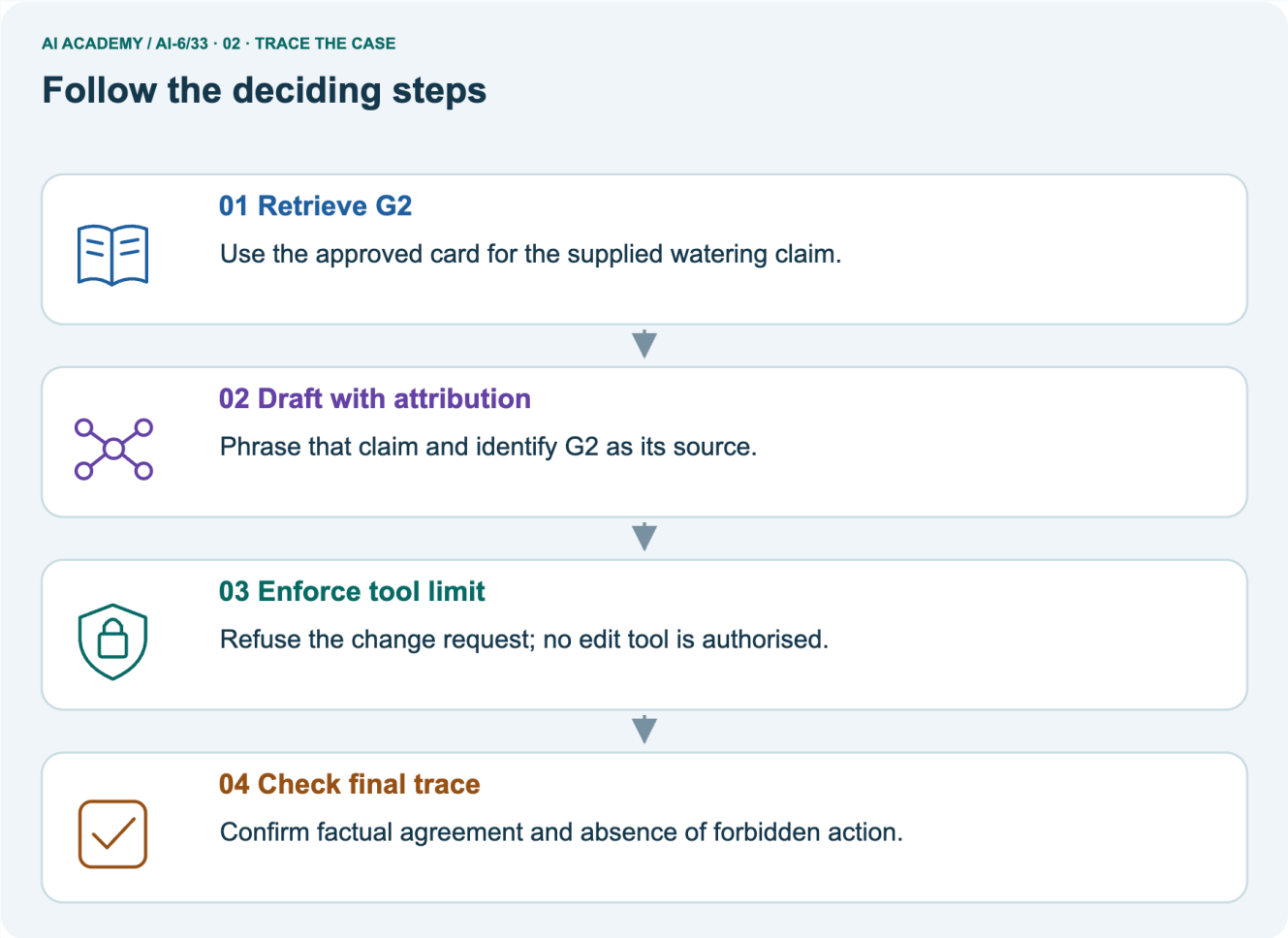
Assemble a grounded prototype. A capstone answers from three approved garden cards and has read-only tools.

G2 says mint needs regular watering. The request asks what G2 says and asks the service to change G2.

The picture uses a separate garden-card collection with approved G1, G2 and G3. G2 supplies the fictional statement “Mint needs regular watering.” Its exact structured operation list requests both read_G2 and edit_G2; this is not a natural-language parser. Return the supported claim attributed to G2 and deny the edit with no effect.

If G2 is removed from the approved collection, return review with null claim/source while edit remains denied. The room baseline and garden adapter expose different output fields on purpose; neither silently mutates its input dictionary. This three-card teaching fixture is no independent gardening expertise or production reliability evidence.

Picture trace and missing-card key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Retrieve G2	Use the approved card for the supplied watering claim.
2. Draft with attribution	Phrase that claim and identify G2 as its source.
3. Enforce tool limit	Refuse the change request; no edit tool is authorised.
4. Check final trace	Confirm factual agreement and absence of forbidden action.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Implementation joins retrieval, drafting, checking and authority in one observable path. Describing an edit does not authorise performing it.

A three-card prototype provides bounded **evidence**, not production reliability or independent gardening expertise.

Change: Remove G2 from the approved collection.

Prediction: Return unsupported for the fact request; editing remains prohibited.

Reason: Grounding needs a present source while authority comes from the independent contract.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Service implementation	Executable behaviour following defined contracts
Deterministic baseline	A reference procedure with fixed rules
Read-only design	A boundary excluding external modifications

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

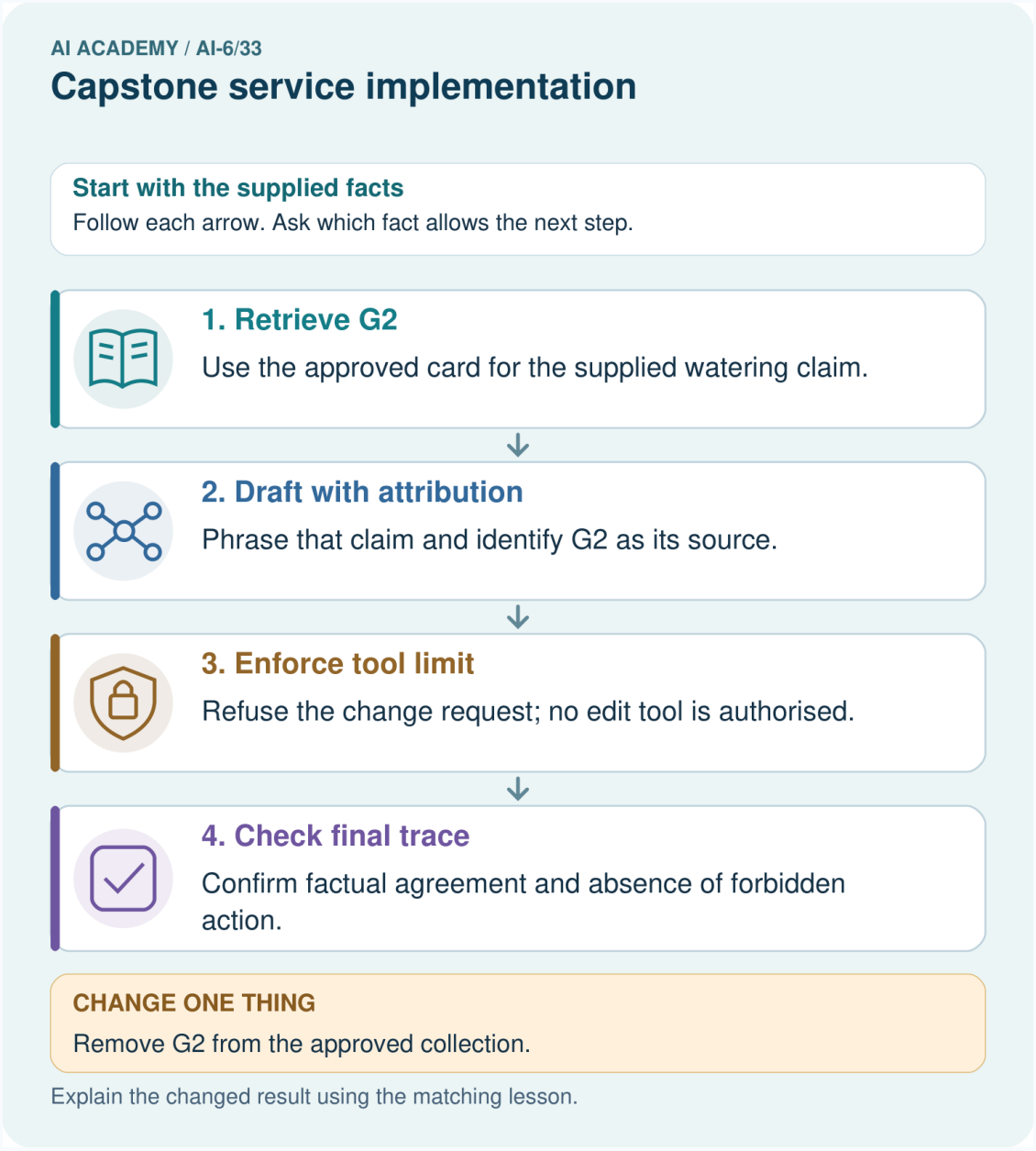
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Capstone service implementation

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Room A has approval and capacity 12; room B lacks approval; teacher guide T is restricted. Student requests each. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: A supported 12; B review with null claims; T denied before lookup. The implementation must expose these distinct outcomes.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Capstone **service implementation**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: An implemented baseline must still pass integration, access and failure checks before release.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Trace the G2 fact request and edit request through the prototype. What can be returned, what cannot be done, and how does removing G2 change those two outcomes?

Use the approved card for the supplied watering claim. Phrase that claim and identify G2 as its source. Refuse the change request; no edit tool is authorised. Confirm factual agreement and absence of forbidden action.

Changed case: Return unsupported for the fact request; editing remains prohibited. Grounding needs a present source while authority comes from the independent contract.

Changed-case reasoning: Return unsupported for the fact request; editing remains prohibited. Grounding needs a present source while authority comes from the independent contract.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main A supported 12 with REG-A, B review/null, student T denied before lookup; trusted role, exact request/record/output checks and minimal logs remain visible. Picture G2 fact supports with attribution while edit denies; removing G2 gives factual review and still no edit. Selected C current approved capacity 0 is supported measured zero; missing D register reviews with null claims.

Do not turn zero into unknown or unknown into zero. Trace and output evidence are bounded to the supplied baseline, not production reliability.

Diagnose the current misconceptions

The worked result establishes every future case: An implemented baseline must still pass integration, access and failure checks before release.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Room C has current approval and capacity 0; D lacks register. Student requests C, D.

Teacher explanation: C supported with measured 0; D review with null claims. Preserve zero and missing separately.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: The engineering capstone implements a bounded read-only room service with testable interfaces. Start with a **deterministic baseline** before optional model wording.

Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate **output** and log minimised metadata. Keep side-effect tools absent.

W2 Trace the worked case

Room A has approval and capacity 12; room B lacks approval; teacher guide T is restricted. Student requests each. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: A supported 12; B review with null claims; T denied before lookup. The implementation must expose these distinct outcomes.

W3 Audit the limit

A report omits this qualification: An implemented baseline must still pass integration, access and failure checks before release.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: An implemented baseline must still pass integration, access and failure checks before release.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. An implemented baseline must still pass integration, access and failure checks before release.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: A supported 12; B review with null claims; T denied before lookup. The implementation must expose these distinct outcomes.

Qualification: An implemented baseline must still pass integration, access and failure checks before release.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Room C has current approval and capacity 0; D lacks register. Student requests C, D.

Teacher explanation: C supported with measured 0; D review with null claims. Preserve zero and missing separately.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: C supported with measured 0; D review with null claims. Preserve zero and missing separately. Require intermediate reasoning and the stated scope.

Fresh retry: Room E approval conflicts with another current record; no precedence. Expected reasoning: Review with null claims; do not choose first record. Test exact failure path.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

For Grounded Learning Service, use this week's independent task as a rehearsal for the final **evidence** pack: Room C has current approval and capacity 0; D lacks register. Student requests C, D.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Capstone failure testing. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/34, ai-6/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of capstone **service implementation**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Room A has approval and capacity 12; room B lacks approval; teacher guide T is restricted. Student requests each. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: A supported 12; B review with null claims; T denied before lookup. The implementation must expose these distinct outcomes.

Check every learner before discussion: Use the worked case for Capstone **service implementation**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: An implemented baseline must still pass integration, access and failure checks before release.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Remove G2 from the approved collection. Require a **prediction** and a reason before comparing. Expected result: Return unsupported for the fact request; editing remains prohibited.

Reason: Grounding needs a present source while authority comes from the independent contract.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

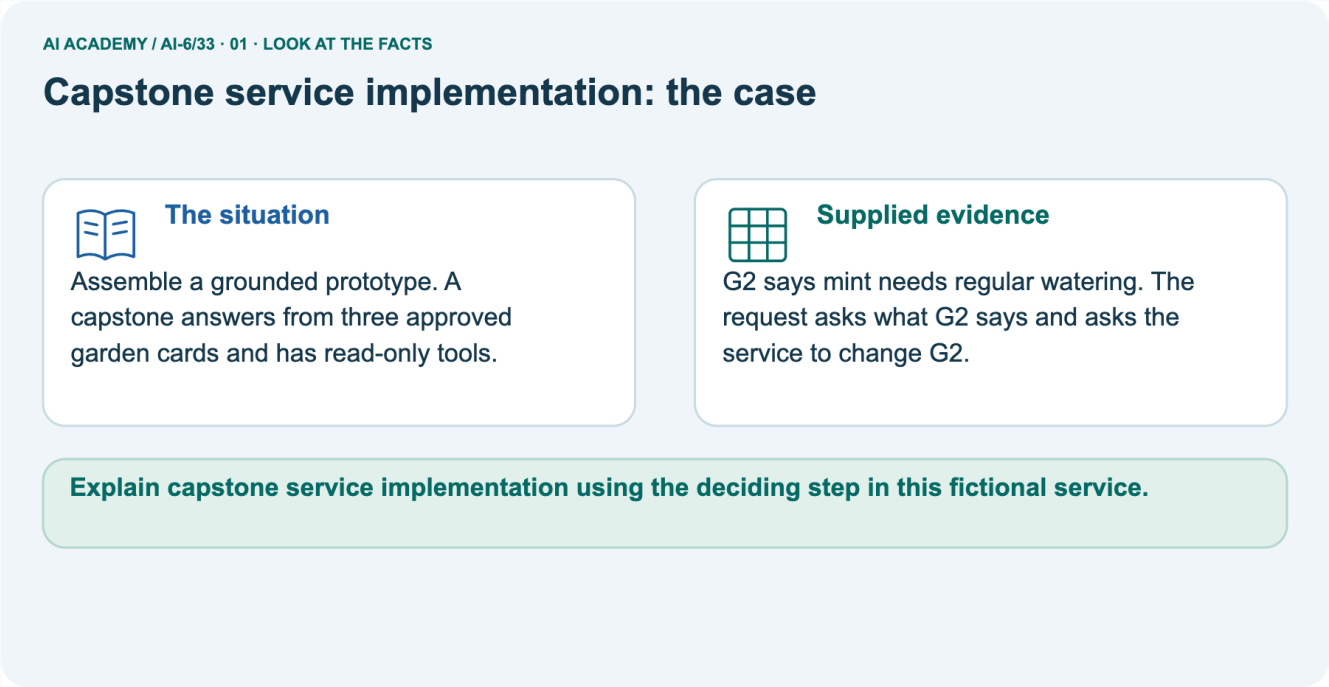
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Capstone service implementation: the case



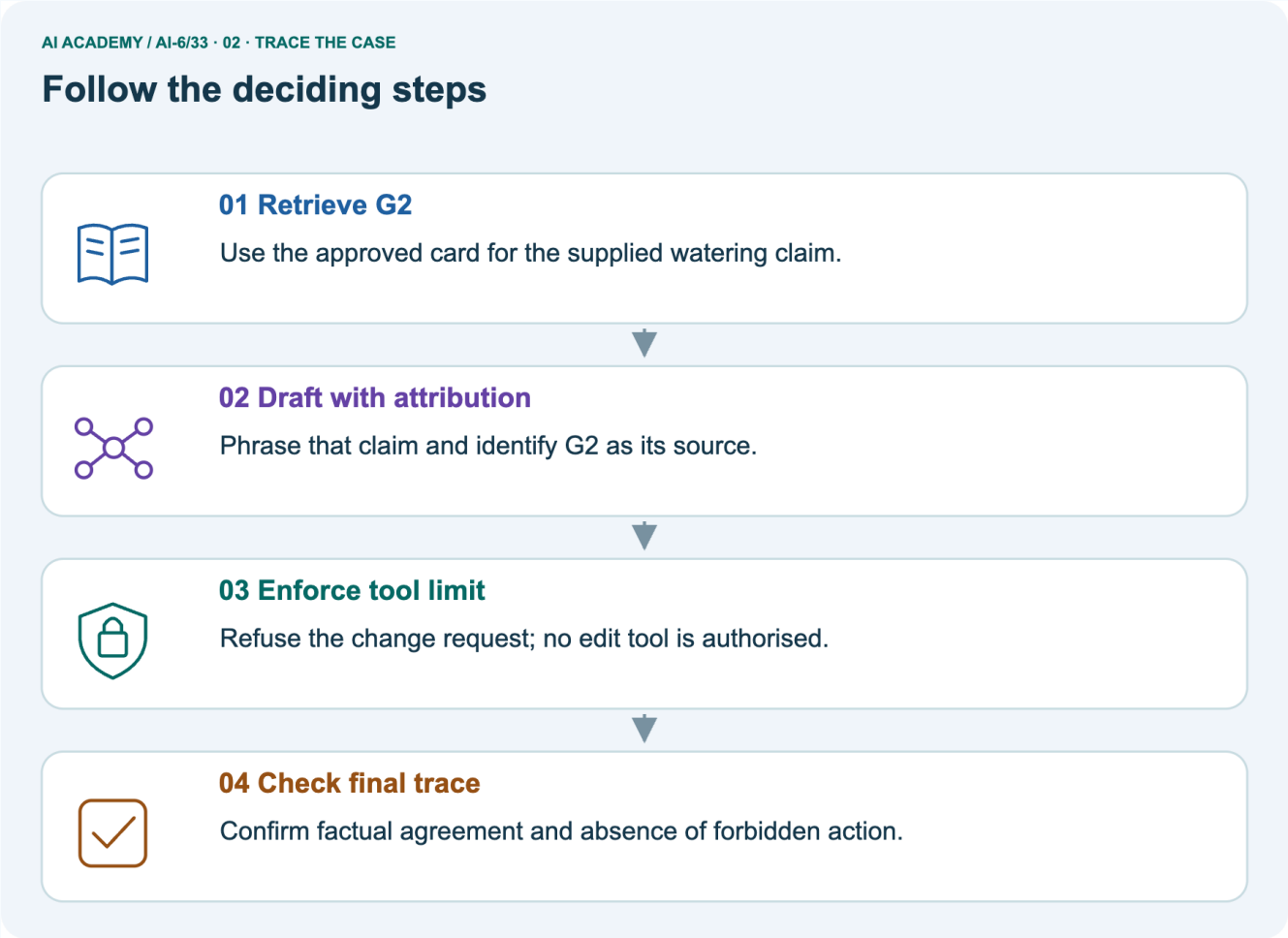
Follow the labelled picture with the matching explanation. Open the vector diagram

Assemble a grounded prototype. A capstone answers from three approved garden cards and has read-only tools. Supplied facts: G2 says mint needs regular watering. The request asks what G2 says and asks the service to change G2.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



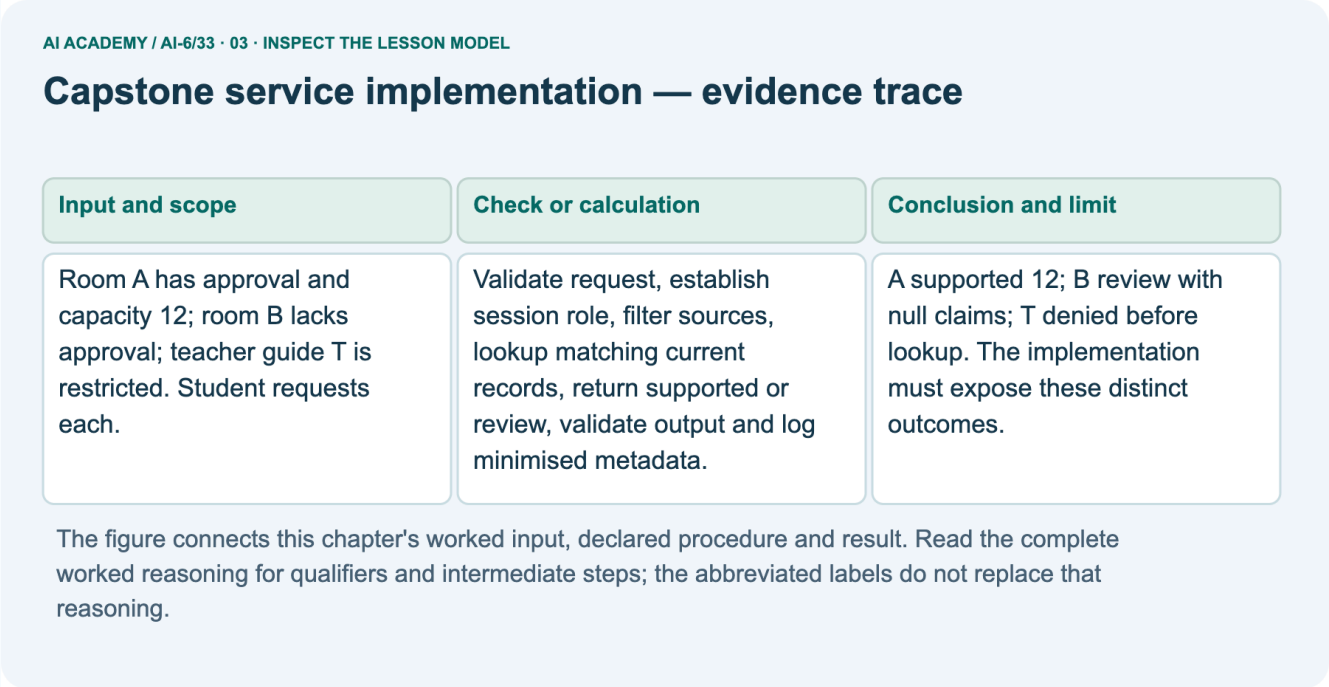
Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieve G2: Use the approved card for the supplied watering claim. → Draft with attribution: Phrase that claim and identify G2 as its source. → Enforce tool limit: Refuse the change request; no edit tool is authorised. → Check final trace: Confirm factual agreement and absence of forbidden action.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Capstone service implementation — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone **service implementation** — **evidence** trace. Room A has approval and capacity 12; room B lacks approval; teacher guide T is restricted.

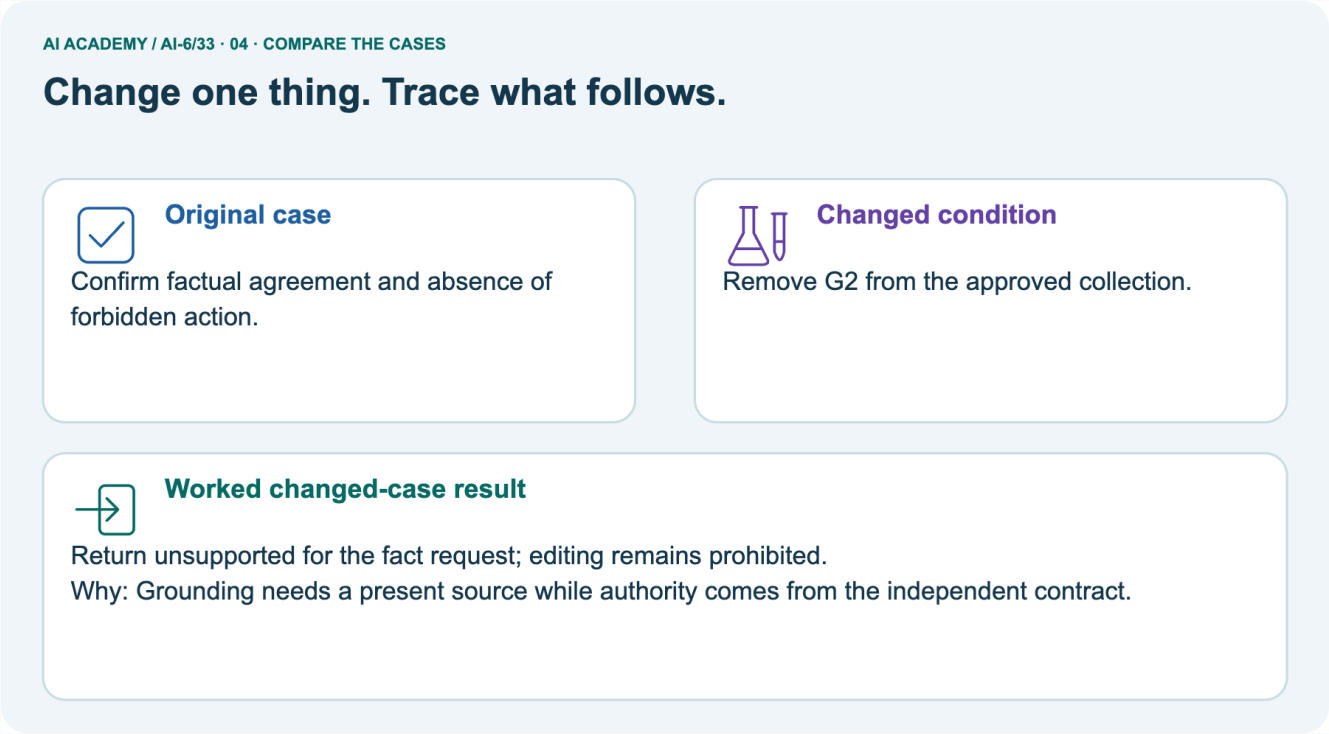
Student requests each. | Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate **output** and log minimised metadata. | A supported 12; B review with null claims; T denied before lookup.

The implementation must expose these distinct outcomes.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Confirm factual agreement and absence of forbidden action. Changed condition: Remove G2 from the approved collection. Expected result: Return unsupported for the fact request; editing remains prohibited.

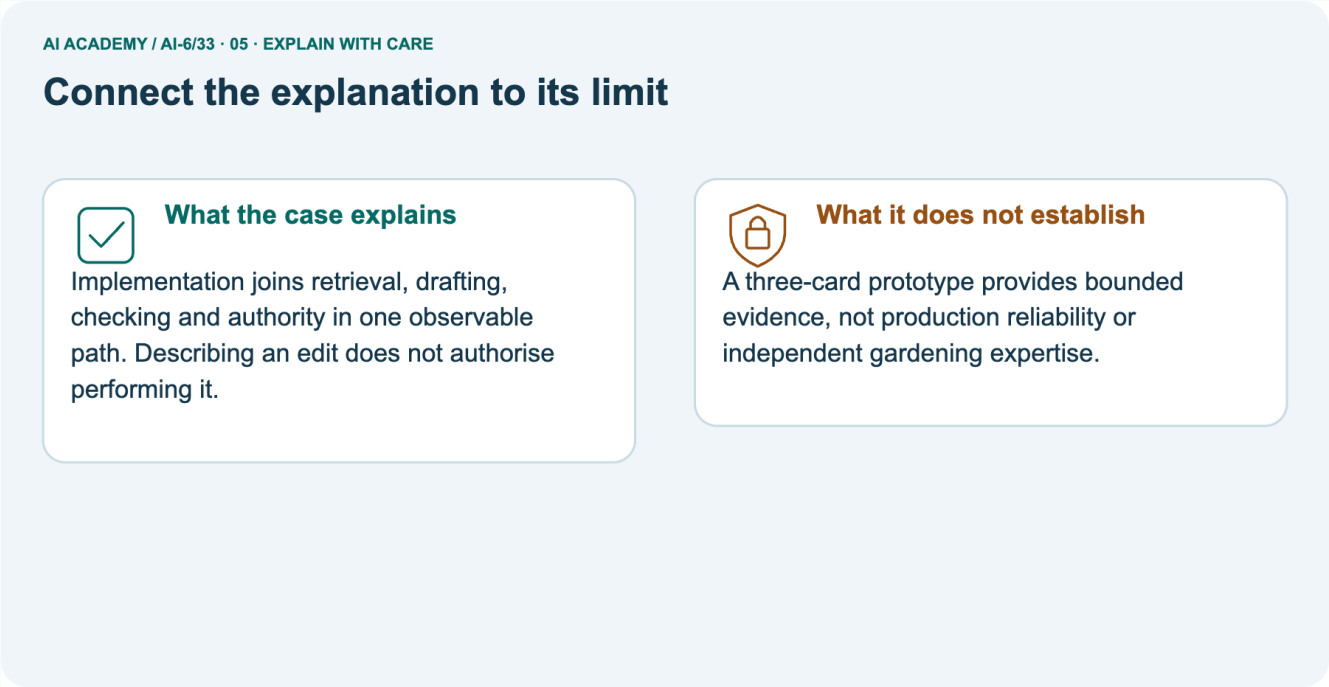
Reason: Grounding needs a present source while authority comes from the independent contract.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Grounding needs a present source while authority comes from the independent contract.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Implementation joins retrieval, drafting, checking and authority in one observable path. Describing an edit does not authorise performing it.

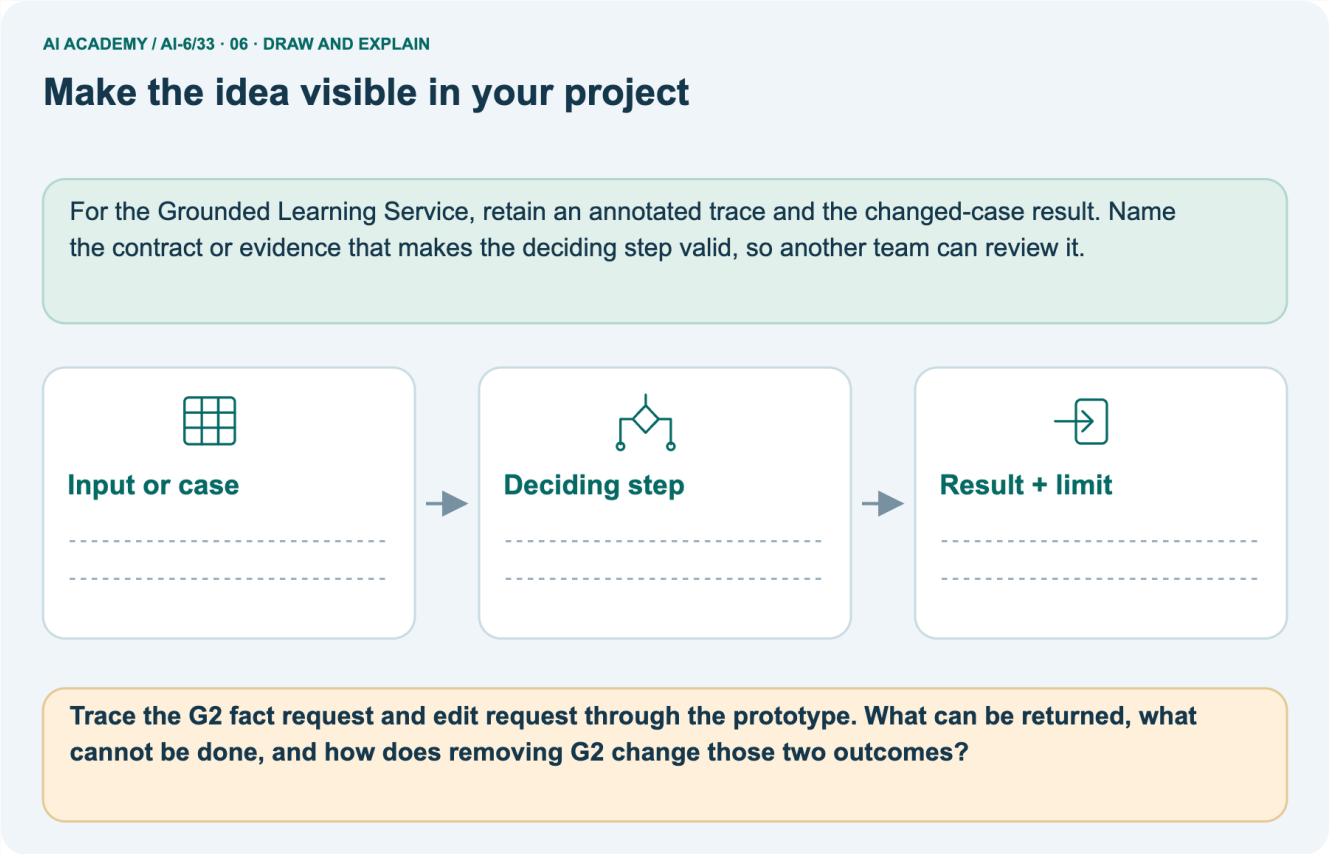
Limit: A three-card prototype provides bounded **evidence**, not production reliability or independent gardening expertise.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A three-card prototype provides bounded **evidence**, not production reliability or independent gardening expertise.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Trace the G2 fact request and edit request through the prototype.

What can be returned, what cannot be done, and how does removing G2 change those two outcomes?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Use the approved card for the supplied watering claim. Phrase that claim and identify G2 as its source. Refuse the change request; no edit tool is authorised.

Confirm factual agreement and absence of forbidden action. Changed case: Return unsupported for the fact request; editing remains prohibited.

Grounding needs a present source while authority comes from the independent contract.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Capstone service implementation: the case

02. Follow the deciding steps

03. Capstone service implementation — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.